

$$\frac{d}{s} = \frac{1}{5}$$

Q1. A 100-km long cable is carrying data at a rate of 2.048 Mbps. The propagation speed in the cable is $2/3$ the speed of light ($c = 3 \times 10^8$ m/sec). How many bits fit in the cable?

a. If the unit exchanged at the data link level is called a frame and the unit exchanged at the network level is called a packet, do frames encapsulate packets or do packets encapsulate frames? Explain your answer.
b. For the below mentioned services mention the name of the corresponding layer of the OSI model.

- Network I. Router selection
- Network II. Define frame data link
- Physical III. Transmission of bit stream across the physical media
- Physical IV. Flow control
- Physical V. Interface to transmission media (physical)
- Application VI. Communicates directly with user's application program
- Application VII. Provide user service such as e-mail and file transfer

We would like to transmit the following data bits using the hamming code with even parity.

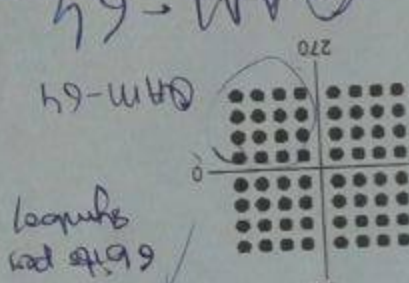
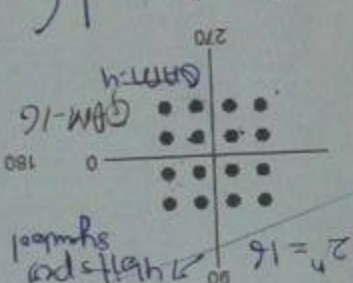
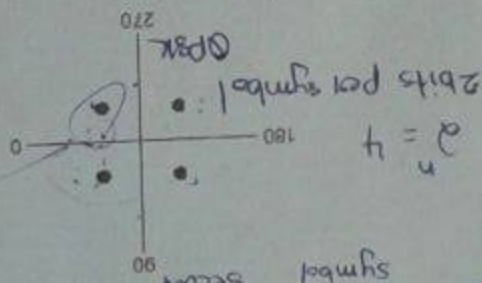
10101111

$m=8, n=9$

Determine the number of check bit required and calculate their value. Clearly identify the data bits contribute to the calculation of each check bit. Give the binary value of the codeword after encoding.

For the following constellation patterns, what type of modulation is being used? Determine the number of bits that can be sent per signal element (symbol). If the symbol rate is 4800 symbols per second, determine the bit rate for each constellation pattern?

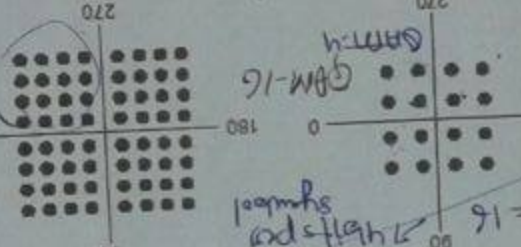
$$\text{bit rate} = \frac{\text{bits}}{\text{symbol}} \times \frac{\text{symbol}}{\text{second}}$$



QAM-16
 $2^4 = 16$
 $2^6 = 64$

$$\text{bit rate} = 4800 \frac{\text{symbols}}{\text{second}} \times \frac{\text{bits}}{\text{symbol}} = 4800 \times 4 = 19200 \text{ bits/second}$$

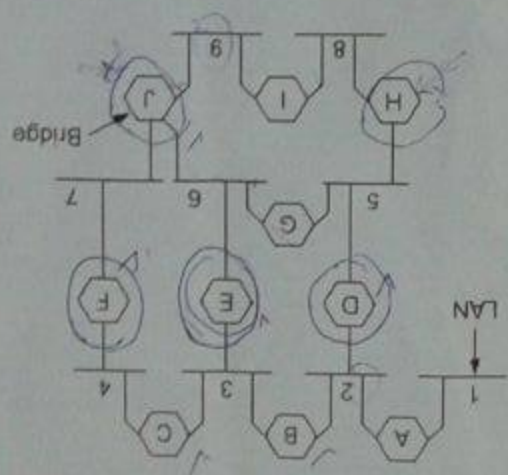
QAM-16
 $2^4 = 16$
 $2^6 = 64$



QAM-64
 $2^6 = 64$
 $2^8 = 256$

Q.4

(a) One consequence of using a spanning tree to forward frames in an extended LAN is that some bridges may not participate at all in forwarding frames. Redraw the following figure and identify such bridges and their links. Is there any reason for keeping these bridges, even though they are not used for forwarding?



(b) Sketch the Manchester encoding on a classic Ethernet for the bit stream 0001110101.

[10]

(a) Assume we need to design a Go-Back-N sliding window protocol for a network in which the bandwidth is 100 Mbps and the average distance between sender and receiver is 10,000 km. Assume the average packet size is 10^5 bits and the propagation speed in the media is 2×10^8 m/s. Find the size of the sender and receiver windows for 100% link utilization, the number of bits in the sequence number, and an appropriate timeout value for the timer.

[10]

(b) A high quality speech signal has a bandwidth of 8 kHz. Suppose that the speech signal is to be quantized and then transmitted over a 28.8 kbps modem. What is the signal to noise ratio (in dB) of the received signal? What modem speed is needed if we require an SNR of 40 dB?

[10]



Course Name:	Data Communication & Networks	Course Code:	EE 317
Program:	Electrical Engineering	Semester:	Spring 2017
Duration:	180 Minutes	Total Marks:	100
Paper Date:	May 15, 2017	Weight:	50%
Section:	ATL	Page(s):	2
Exam Type:	Final		

Student Name: MISHAB GHABRI Roll No. 14-UG1 Section: EE-2

Instruction/Notes: 1. Show all the work for every question. 2. Attempt all parts of a question together.

Q.1

- (a) For the URL, <http://www.cs.washington.edu/index.html>, identify and name the different parts of the URL. List the steps a web browser will carry out in order to fetch the web page pointed to by this URL. Assume that the web page includes embedded URLs and embedded images?

[CLO 7][10]

- (b) A portion of DNS database for `cs.vu.nl` is given below. Explain the meaning of each line in the database.

1. cs.vu.nl	86400	IN	SOA	star boss (95771,7200,7200,24900,8640)
2. cs.vu.nl	86400	IN	MX	1 zephyr
3. cs.vu.nl	86400	IN	NS	star
4. star	86400	IN	A	130.37.56.205
5. zephyr	86400	IN	A	130.37.20.10
6. www	86400	IN	CNAME	star.cs.vu.nl

Internet

Q.2

- (a) Both UDP and TCP use port numbers to identify the destination entity when delivering a message. Give two reasons why these protocols invented a new abstract ID (port numbers), instead of using process IDs, which already existed when these protocols were designed.

[CLO 6][10]

- (b) A process on host 1 has been assigned port `p`, and a process on host 2 has been assigned port `q`. Is it possible for there to be two or more TCP connections between these two ports at the same time?

[CLO 6][5]

- (c) The maximum payload of a TCP segment is 65,495 bytes. Why was such a strange number chosen?

[CLO 6][5]

- (a) A large number of consecutive IP addresses are available starting at 204.18.0.0. Suppose that four organizations, A, B, C, and D, request 2000, 1000, 2000, and 4000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the `w.x.y.z/s` notation.

[CLO 5][10]

OSI model

- ① Application Provide access for end user/email, file transfer/HTTP
- ② Presentation
- ③ Session
- ④ Transport flow control/reliable process to process message delivery.
- ⑤ Network Route determine.
- ⑥ Data link Refine frames b/w adjacent nodes.
- ⑦ Physical Interface to transmission media/Transmission of a bit stream over physical medium

TCP/IP mode

- ① Application layer Provide service for end users
- ② Transport layer / host-to-host UDP/TCP
- ③ Internet layer IP/ICMP
- ④ Network layer Route determine
- ⑤ Physical Connect to transmission media/Transforming bits to electric/magnetic signals

$$10 \log_{10} \frac{S}{N} \rightarrow \text{SNR in dB}$$

$$\frac{1}{10} \log_{10} \frac{S}{N}$$



Course:	Data Communication & Networks	Course Code:	EE317
Program:	BS(Electrical Engineering)	Semester:	Fall 2016
Duration:	3 hrs	Total Marks:	100
Paper Date:	15-December-16	Weight:	50%
Section:	ALL	Page(s):	2
Exam:	FINAL	Roll No:	13L-4446

Instruction/Notes: Show all working/ steps required to get the final answer. Attempt all parts of a question together.

You are entrusted with the design of a network to interconnect a set of geographically distributed hosts within your corporation. You have decided to use a copper-based, switched network with an 8-point QAM encoding. The technology uses 5MHz signal bandwidth and 511 SNR at transmitter. What link bandwidth is possible according to Shannon's Law for this network? What modulation rate (baud) is necessary to obtain this bandwidth?

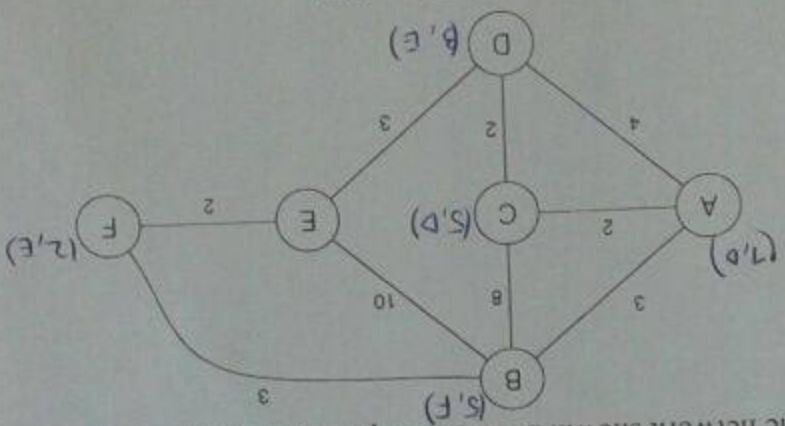
Frames of 1000 bits are sent over a 1-Mbps channel using a geostationary satellite whose propagation time from the earth is 270 msec. Acknowledgements are always piggybacked onto data frames. The headers are very short. Three-bit sequence numbers are used. What is the maximum achievable channel utilization for Stop-and-wait and Selective Repeat?

The USB protocol uses a CRC with the polynomial $x^5 + x^2 + 1$.

Calculate the CRC of the bit string 10011001. Assume bit string 101110111000 is received (using the same generator G as in a). Determine if error could be detected or not.

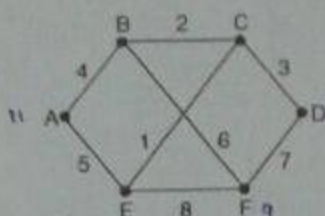
Consider a frame consisting of four bits. Assume that the probability of error is 10^{-3} , which is independent for each bit. What is the probability that the frame is received correctly?

Suppose that routing information is exchanged using link state routing protocol and the shortest paths are computed using Dijkstra's algorithm. Find the shortest path from E to all other nodes in the network shown. Show the steps in your computation.



What are distance vectors in nodes E, D, and C? Note: you do not have to run the distance vector algorithm; compute distance vectors by inspection.

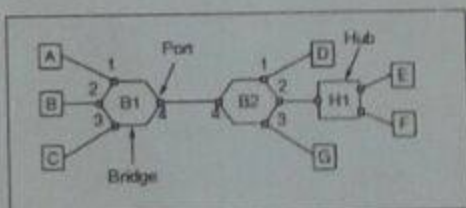
- (b) Consider the network given below. Distance vector routing is used, and the following vectors have just come in to router C: from B: (5, 0, 8, 12, 6, 2); from D: (16, 12, 6, 0, 9, 10); and from E: (7, 6, 3, 9, 0, 4). The cost of the links from C to B, D, and E, are 6, 3, and 5, respectively. What is C's new routing table? Give both the outgoing line to use and the cost. [CLO 5][10]



- Q.4 (a) A 3000-km-long T1 trunk (1.544 Mbps) is used to transmit 64-byte frames using Go-Back-N protocol. If the propagation speed is 6 $\mu\text{sec/km}$, how many bits should the sequence numbers be? [CLO 2][10]

- (b) Consider the extended LAN connected using bridges B1 and B2 as shown in the figure below. Suppose the hash tables in the two bridges are empty. List all ports on which a packet will be forwarded for the following sequence of data transmissions: [CLO 3][10]

- A sends a packet to C.
- E sends a packet to F.
- F sends a packet to E.
- G sends a packet to E.
- D sends a packet to A.
- B sends a packet to F.



234	123
123	134
x	x
x	2
1	4
134	1,2,3

- Q.5 (a) A modem constellation diagram has data points at the following coordinates: (1, 1), (1, -1), (-1, 1), and (-1, -1). How many bps can a modem with these parameters achieve at 1200 symbols/second? 2400 [CLO 1][10]

- (b) Ten signals, each requiring 4000 Hz, are multiplexed onto a single channel using FDM. What is the minimum bandwidth required for the multiplexed channel? Assume that the guard bands are 400 Hz wide. 44000 [CLO 1][10]

EE317, Data Communication & Computer Networks (EE)

Mid Term - I (Fall 2016)

Maximum Marks: 30

Instructor: Ms. Beenish Fatima

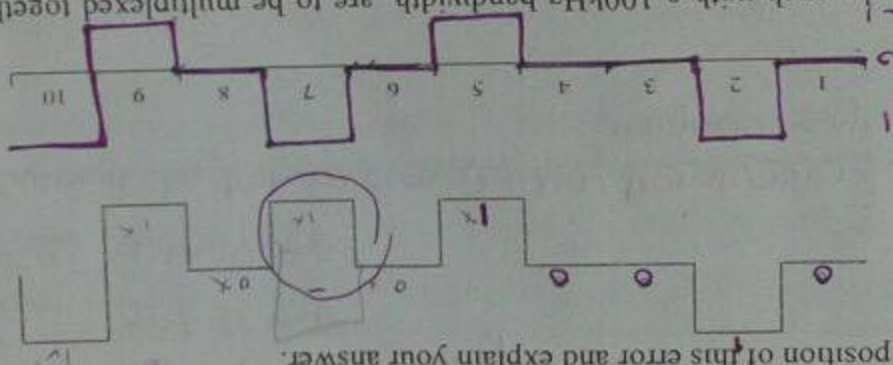
Dated: 21st September, 2016

Total Time: 60mins

Assume that a TV picture is to be transmitted over a channel whose spectrum is between 3MHz and 4MHz. The signal-to-noise ratio for this channel is 24dB. Find the appropriate bit rate and signaling levels required?

Question # 1 (10 marks)

The bipolar-AMI waveform representing the binary sequence 0100101011 is transmitted over a noisy channel. The received waveform shown in the figure below contains a single error. Locate the position of this error and explain your answer.



Five channels, each with a 100kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10kHz between the channels to prevent interference? Also draw a figure to show the multiplexed channels.

$$B = (5 \times 100k) + (4 \times 10k)$$

Question # 3 (6 + 4 = 10 marks)

Match the following to one or more layers of the TCP/IP protocol suite:

- i. Route determination *Network*
- ii. Connection to transmission media *physical*
- iii. Providing services for the end user *Application*
- iv. Creating user datagrams (UDP) *Transport*
- v. Transforming bits to electromagnetic signals *physical*
- vi. Handling frames between adjacent nodes *Data Link*

b) Briefly discuss the consequences of connection failure for a network consisting of 7 devices arranged in each of the following topologies: mesh, star, bus and ring.



Course:	Data Communication & Networks
Program:	BS(Electrical Engineering)
Duration:	60 Minutes
Paper Date:	10-Nov-16
Section:	ALL
Exam:	Midterm-II
Course Code:	EE317
Semester:	Fall 2016
Total Marks:	50
Weight:	15%
Page(s):	1
Roll No:	

Instruction/Notes: Show all working/ steps required to get the final answer

Q.1 The following Hamming coded (single bit correction) string was received: 010101010111. Is there an error? If yes, in which position?

A 3000-km long T1 trunk with data rate of 1.544Mbps is used to transmit 64-byte frames using Go-Back-N protocol. If the propagation speed is 6µsec/km, how many bits should the sequence number be?

Q.2 A 64-kbps pure ALOHA channel is to be shared among a number of stations, each sending a 1k bit frame, on average one frame every 10 seconds. (Assume the sender can buffer frames to handle variations in successful sending). What is the maximum number of stations this network can support? Hint: A pure ALOHA channel has a maximum capacity of 18.4%.

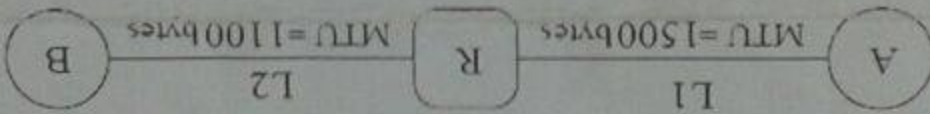
Q.3 Consider building a CSMA/CD network running at 1 Gbps over a 1 km cable. There are no repeaters. The signal speed in the cable is 200000 km/sec. What is the minimum frame size (in bits)?

b) What is hidden node problem and exposed node problem in 802.11 networks? How is this issue resolved? Use diagrams to facilitate your answer.

Q4. A large number of consecutive IP addresses are available starting at 198.24.0.0. Three organizations A, B and C request 2048, 3096 and 1024 addresses respectively in the given order. For each of these, what is the first IP assigned, last IP assigned and the mask in w.x.y.z/s notation? How many addresses are remaining? Also state the advantages of classless addressing.

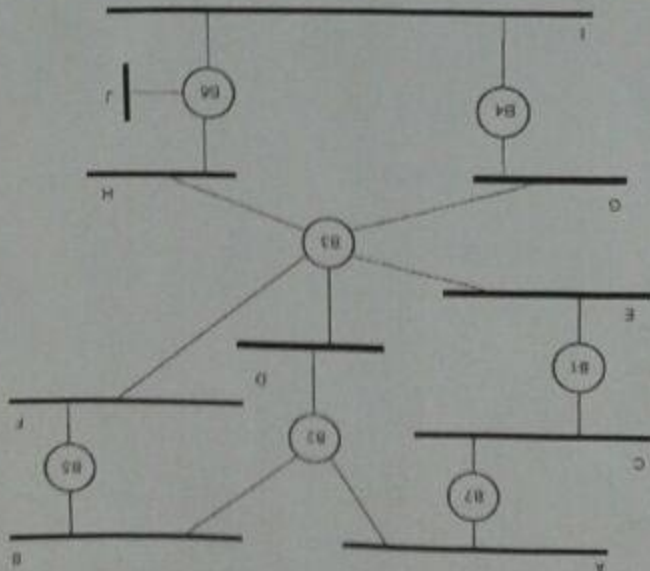
(20)

Q5. Show all the fragments that will be created when the datagram is sent first from A to R over L1 and then from R to B over L2. How many fragments does B receive? Specify values in the fragment header of all fragments?



Q6. The diagram below illustrates Host A sending a datagram of size 4000 bytes (including header) to host B through Router R.

(10)



Q7. Given the Ethernet LAN shown in the figure below, indicate which ports are not selected (and therefore disabled) by the spanning tree algorithm. The hosts in the figure are labeled A-J and the bridges in the figure are labeled B1-B7. Also tell the path that would be used for sending data from Host A to Host H.

(10)



Course:	Data Communication & Networks	Course Code:	EE3 17
Program:	BS (Electrical Engineering)	Semester:	Spring 2017
Duration:	60 Minutes	Total Marks:	40
Paper Date:	20-Feb-17	Weight:	15%
Section:	All	Page(s):	1
Exam:	Midterm-1	Roll No:	

Solve All Questions. Attempt all parts of a question together.

Q.1 During the negotiation period, the transmitter and receiver have agreed to use the polynomial 1001 for their CRC code. Suppose that a receiver has received the following frame 11001001010. Should the receiver accept or reject the frame? [10] 10

Q.2 A typical voice-grade telephone line has a bandwidth of 3100Hz and a signal-to-noise ratio of 30dB, which limits the data rate to about 30kbps. This bandwidth is artificially created by a low pass filter on the local loop. To achieve DSL speeds, this low pass filter is removed. To attain DSL speeds of 6 Mbps downstream and 512 kbps upstream, what is the minimum bandwidth required on the local loop? [10]

Q.3 (a) For the bit stream 01011100, sketch the waveform for NRZ, NRZI and Manchester encoding. Assume that the signal level of preceding bit is high. [5] 5

(b) Give one advantage and disadvantage of using Manchester encoding compared to NRZ. [5] 3

Q.4 (a) The following data fragment occurs in the middle of a data stream for which the byte stuffing algorithm is used: A B ESC C ESC FLAG FLAG D [5] 3

What is the output after stuffing? [5] 3

(b) One of your classmates has pointed out that it is wasteful to end each frame with a flag byte and then begin the next one with a second flag byte. One flag byte could do the job as well, and a byte saved is a byte earned. Do you agree? Explain. [5] 3

Q.5

Advantage
Detection of consecutive '1' or '0' is easy in mc but difficult in NRZ

Disadvantage
mc needs twice as much BW as NRZ b/c of the clock

Department of Electrical Engineering
mc may have time delays

Q.1

Suppose that host A is connected to a router R1, R1 is connected to another router, R2, and R2 is connected to host B. Suppose that a TCP message that contains 900 bytes of data and 20 bytes of TCP header is passed to the IP code at host A for delivery to host B. Show the **Total length**, **Identification**, **DF**, **MF**, and **Fragment offset** fields of the IP header in each packet transmitted over the three links. Assume that link A-R1 can support a maximum data link frame size of 1024 bytes including a 14-byte frame header, link R1-R2 can support a maximum data link frame size of 512 bytes, including a 8-byte frame header, and link R2-B can support a maximum data link frame size of 512 bytes including a 14-byte frame header.

[20]

Q.2

(a) A large number of consecutive IP addresses are available starting at 202.67.0.0. Suppose that four organizations, A, B, C, and D, request 1500, 3000, 2000, and 16000 addresses respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation. [10]

(b) A router has the following (CIDR) entries in its routing table: [10]

<u>Address/mask</u>	<u>Next Hop</u>
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Router 1
default	Router 2

For each of the following IP addresses, what does the router do if a packet with that destination address arrives? Show all the calculations if you want any credit for this question.

- (i) 135.46.63.10
- (ii) 135.46.57.14
- (iii) 135.46.52.2
- (iv) 192.53.40.7
- (v) 192.53.56.7

Q.3

A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$. Determine the actual bit stream transmitted. If the third bit from the left is inverted during transmission, show that this error is detected at the receiver. [20]

Q2:

$$C = B \log_2(1 + \text{SNR})$$

$$= 3100 \log_2(1 + 100)$$

$$C = 30.8 \text{ kbps} \approx 30 \text{ kbps}$$

$$C_{\text{down}} = 6 \text{ Mbps} \Rightarrow B_1 = 602 \text{ kHz}$$

$$C_{\text{up}} = 512 \text{ kbps} \Rightarrow B_2 = \frac{C_{\text{up}}}{\log_2(1001)} = 51.4 \text{ kHz}$$

$$B_{\text{min}} = B_1 + B_2$$

Q4(a)

one esc $\longrightarrow$ two esc

one esc one flag $\longrightarrow$ two esc



Course: Data Communication & Networks
Program: BS (Electrical Engineering)
Duration: 60 Minutes
Paper Date: 6-April-17
Section: All
Exam: Midterm-2

Course Code: EE 317
Semester: Spring 2017
Total Marks: 40
Weight: 15%
Page(s): 1
Roll No: 14-4461

Instruction/Notes: Solve All Questions. Attempt all parts of a question together.

N = 4/5

Q.1 (a) A 1-km-long, 10-Mbps CSMA/CD LAN (not 802.3) has a propagation speed of 200 m/μsec. Repeaters are not allowed in this system. Data frames are 256 bits long, including 32 bits of header, checksum, and other overhead. The first bit slot after a successful transmission is reserved for the receiver to send a 32-bit acknowledgement frame. What is the effective data rate, excluding overhead, assuming that there are no collisions?

Q.1 (b) An IP packet to be transmitted by Ethernet is 55 bytes long, including all its headers. If LLC is not in use, is padding needed in the Ethernet frame, and if so, how many bytes? [CLO 3] [5]

Q.2 (a) Suppose that an 11-Mbps 802.11b LAN is transmitting 64-byte frames back-to-back over a radio channel with a bit error rate of 10^{-7} . How many frames per second will be damaged on average? (10) [CLO 4] [10]

Q.2 (b) In 802.11 frame header there is 2-byte duration field. Explain its purpose. [CLO 4] [5]

Q.3 (a) Datagram networks route each packet as a separate unit, independent of all others. Virtual-circuit networks do not have to do this, since each data packet follows a predetermined route. Does this observation mean that virtual-circuit networks do not need the capability to route isolated packets from an arbitrary source to an arbitrary destination? Explain your answer. [CLO 5] [5]

(b) Assuming that all routers and hosts are working properly and that all software in both is free of all errors, is there any chance, however small, that a packet will be delivered to the wrong destination? [CLO 5] [5]

Q.4 (b) minimum ethernet frame is 64 bytes
header field occupy is 18 bytes
packet is 55 bytes
Hence total frame size = 55 + 18 = 73 bytes which exceeds 64-byte min so 'No'

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padding

Q.2 (a) $p = 10^{-7}$ (corrupt)
 $(1-p)^{512} = 0.999 \approx 1$ (without corrupt rate)
multiply ≈ 1 frame damaged per second

fraction damaged $= 5 \times 10^{-5}$
 $\frac{\text{frames}}{\text{sec}} = \frac{512}{11 \times 10^6} \approx 21,484$

$\frac{\text{bits/sec}}{\text{bits/frame}} = \frac{\text{frames/sec}}{\text{frames/sec}}$

Q2(a)

Yes we have to route, just for establishing connection for the first time. After that the packets know their route.

(b) Hardware, Software & routers, Yes, when noise burst changes the destination address & virtual circuit ID

in Data Gram Network

- Q2(b) (i) Meaning of field
- ② Hello duration & transmission & acknowledgment
 - ③ Job broadcast no go to back receiver NAB known go and remain silent.
 - ④ We are avoiding collision in whole scenario

① (a) : frame time = $\frac{\text{frame size}}{\text{data rate}} = \frac{256 \text{ bits}}{10 \text{ Mbps}} = 25.6 \times 10^{-6}$

data actual bit (excluding overhead) $256 - 32 = 224$

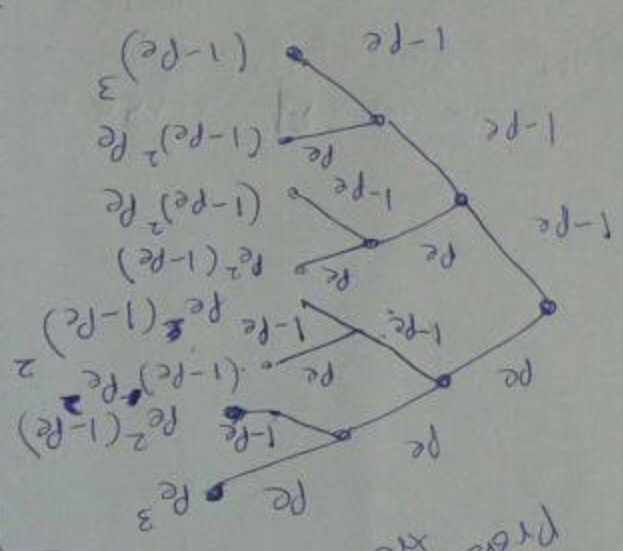
propagative time = $\frac{\text{distance}}{\text{speed}} = \frac{1000}{200 \text{ m/us}} = 5 \text{ us}$

$25.6 \text{ us} + 5 \text{ us}$

$\frac{32 \text{ bits}}{10 \times 10^6 \text{ bits/s}} = 3.2 \text{ us}$

$(3.2 + 5 + 5 + 25.6) \text{ us}$

Probability tree



$pe^3 + 3pe^2(1-pe) + 3pe(1-pe)^2 + (1-pe)^3 = 1$

one frame damage = $p_0 = 1 - (1-pe)^3$ [fraction damage]

probability is $= 1 - (1-10^{-7})^{512} = 5.11 \times 10^{-5}$

→ Q.1

Consider an error-free 64-kbps satellite channel used to send 512-byte data frames in one direction, with very short acknowledgements coming back the other way. What is the maximum throughput for window sizes of 1 and 9? The one way propagation time is 270 msec.

[10]

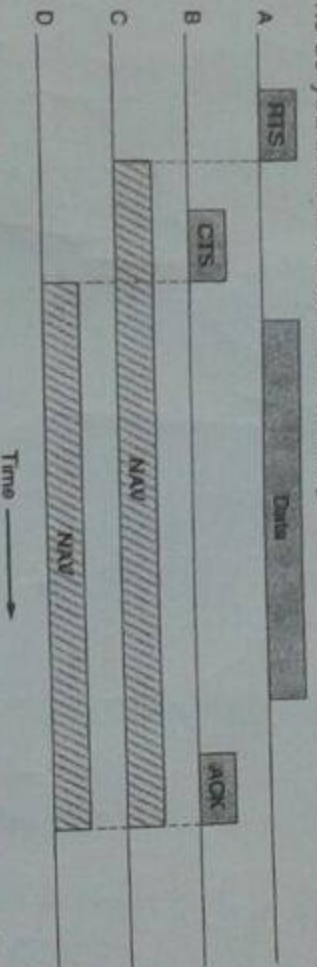
Q.2

(a) Discuss the **binary exponential backoff** algorithm. Why it has been chosen in 802.3 LAN to handle the collisions. This algorithm uses 51.2 μ sec contention slots, why? [5]
 (b) An IP packet to be transmitted by an 802.3 Ethernet is 40 bytes long, including all its headers. If LLC is not in use, padding is needed in the Ethernet frame, and if so how many bytes? [5]

→ Q.3

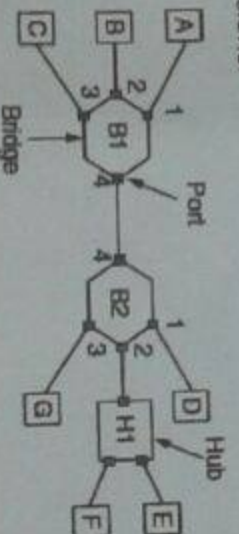
(a) If the probability of any bit being in error is 10^{-4} , calculate the probability of 512-bit 802.3 frame being received without error. [5]

(a) In the following figure four stations, A, B, C, and D, are shown. Which of the last two stations do you think is closest to A and why? [5]



Q.4

Consider the extended LAN connected using bridges B1 and B2. Suppose the hash tables in the two bridges are empty. List all ports on which a packet will be forwarded for the following sequence of data transmissions: [10]



- A sends a packet to C.
- E sends a packet to F.
- F sends a packet to E.
- G sends a packet to E.
- D sends a packet to A.
- B sends a packet to F.

National University of Computer and Emerging Sciences

EE 317 Data Communication & Networks (EE-2)

Spring 2018

Quiz no.4

Total Marks: 10

Time allowed: 10mins

Roll num: L15-4522

CLO 06

Date: 02-05-2018

a) When a TCP packet arrives at a host, in order to direct the segment to the appropriate socket, the operating system's network stack uses the following fields: (2)

- ☒ i. Transport protocol number
- ii. Destination IP address
- iii. Source port number
- ☒ iv. Destination port number
- v. Destination MAC address

b) What is the purpose of TCP/UDP port numbers? (1)

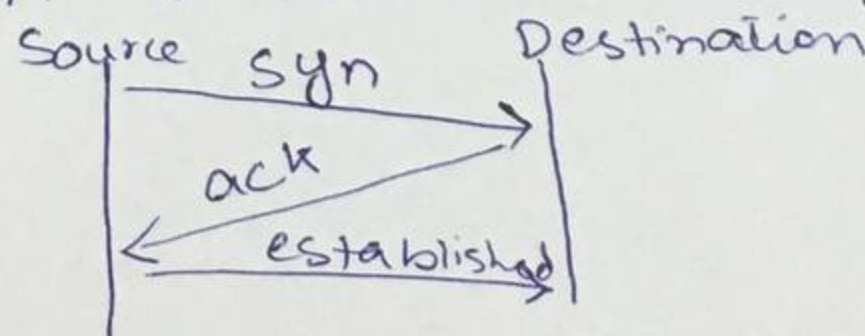
- vi. To indicate the beginning of a three-way handshake
- vii. To reassemble the segments into the correct order
- viii. To identify the numbers of the data packets that can be sent without acknowledgment
- ☒ ix. To choose among multiple processes running on a host

c) Which of the following system calls results in the sending of SYN packets? (4)

- (A) socket
- (B) bind
- (C) listen
- ☒ (D) connect

Justify your answer and explain how a connection is established in TCP.

TCP connection is established by 3 way handshaking-



d) Consider a TCP connection in a state where there are no outstanding ACKs. The sender sends two segments back to back. The sequence numbers of the first and second segments are 230 and 290 respectively. The first segment was lost, but the second segment was received correctly by the receiver. Let X be the amount of data carried in the first segment (in bytes), and Y be the ACK number sent by the receiver. (3)

The values of X and Y (in that order) are

- ☒ (A) 60 and 290
- ☒ (B) 230 and 291
- (C) 60 and 231
- (D) 60 and 230

Receiver will send the acknowledgement of 291 meaning that it has received 290th byte and now is waiting for 291th byte

a) When a TCP packet arrives at a host, in order to direct the segment to the appropriate socket, the operating system's network stack uses the following fields: (2)

- Transport protocol number
- Destination IP address
- Source port number
- Destination port number
- Destination MAC address

1 1/2

b) What is the purpose of TCP/UDP port numbers? (1)

- To indicate the beginning of a three-way handshake
- To reassemble the segments into the correct order
- To identify the numbers of the data packets that can be sent without acknowledgment
- To choose among multiple processes running on a host

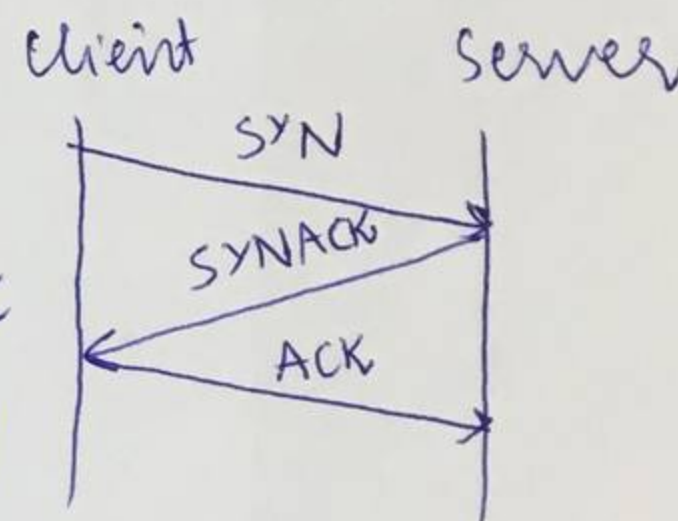
c) Which of the following system calls results in the sending of SYN packets? (4)

- socket
- bind
- listen
- connect

3 1/2

Justify your answer and explain how a connection is established in TCP.

Client send request to establish connection to server with Flag SYN. Server sends SYNACK flag and client send ACK flag. This three way handshaking establishes connection. Now data transfer starts.

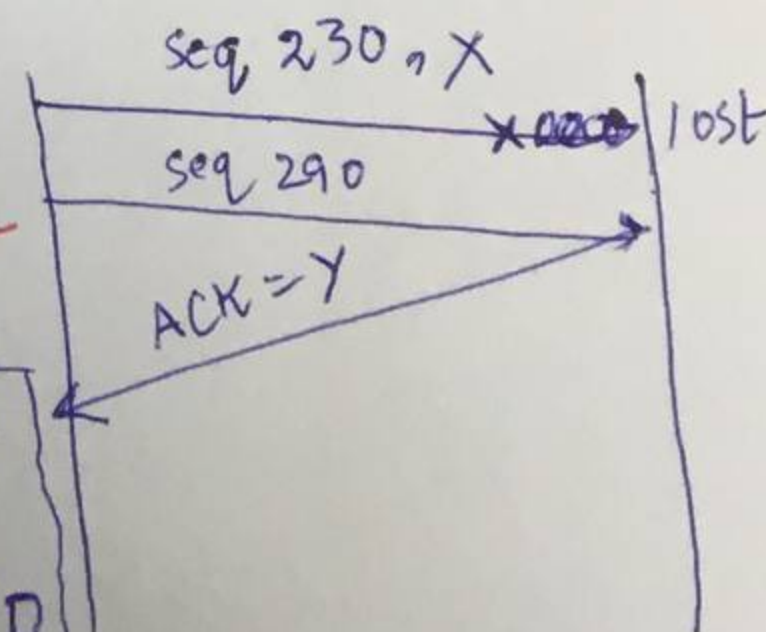


d) Consider a TCP connection in a state where there are no outstanding ACKs. The sender sends two segments back to back. The sequence numbers of the first and second segments are 230 and 290 respectively. The first segment was lost, but the second segment was received correctly by the receiver. Let X be the amount of data carried in the first segment (in bytes), and Y be the ACK number sent by the receiver. (3)

The values of X and Y (in that order) are

- 60 and 290
- 230 and 291
- 60 and 231
- 60 and 230

3



• Seq of next packet

$$X = 290 - 230 = 60$$

$$Y = 230$$

⇒ Server was ready to receive 230 seq# so when out of order seq 290 came of 230 was sent again

Question No. 3 (For Sections EE-1 and EE-3 only)

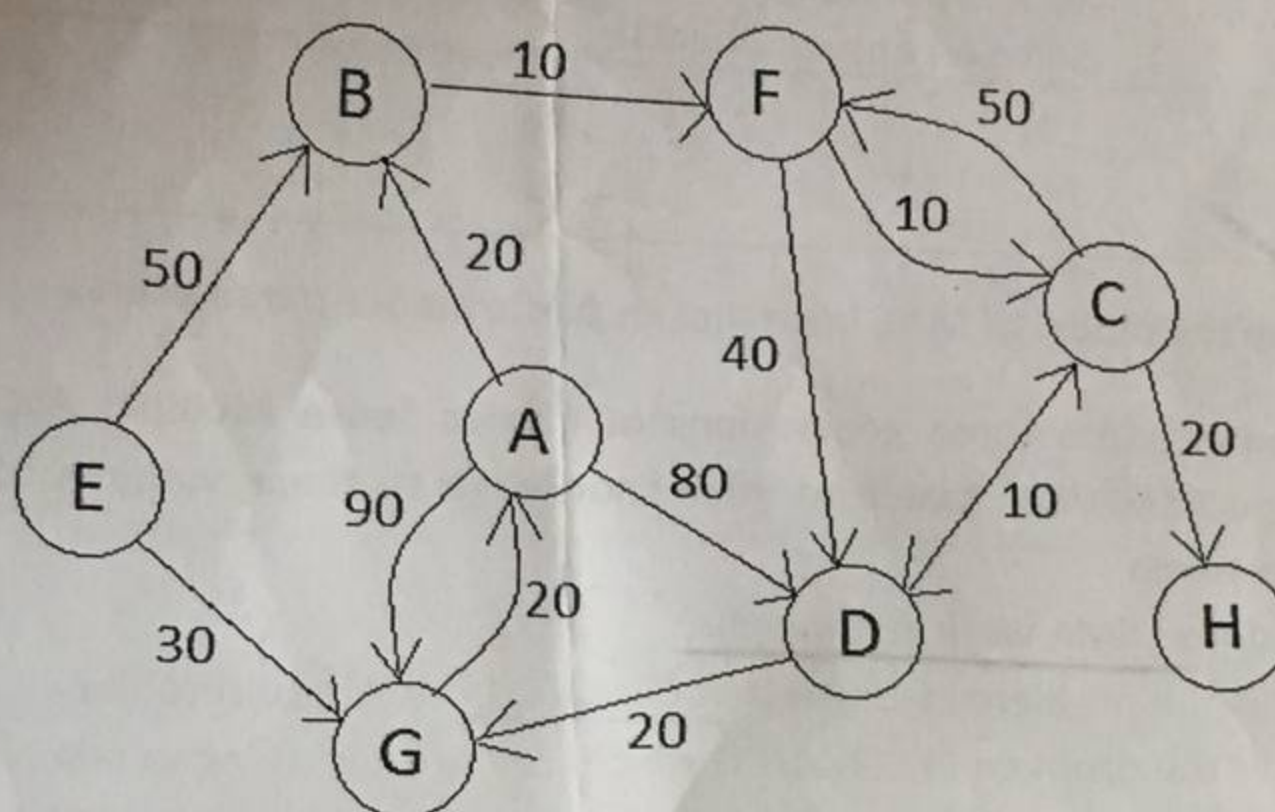
Marks: 10

CLO 05 Evaluate network layer protocols and their impact on network performance.

Using Dijkstra's Algorithm, compute the shortest paths from node A to all other nodes of the network. You must write complete equations (plus put values in equations) for all iterations.

Clearly indicate which distances are updated in each iteration.

[Note: This is a directed graph with some unidirectional links, some bidirectional links; some links with symmetric costs and some with asymmetric costs]



Question No. 3 (For Section EE-2 only)

Marks: 10

CLO 05 Evaluate network layer protocols and their impact on network performance.

- a) Consider the IP address 184.86.92.182 to solve the following questions.
 - i. Suppose that we were using class-based addressing. What type of network would this IP address be a part of?
 - ii. If the network administrator decided to break this network into 8 equal subnets, how many hosts can be on this subnet? What would be the subnet mask of the subnet to which this IP address belongs?
 - iii. What is the range of IP addresses for this subnet? In other words, what is the starting IP address and ending IP address of this subnet?
 - iv. Now suppose instead that we are using CIDR addressing instead of Class-based addressing and subnets. What would be the length of CIDR prefix for the physical network in part iii) to which the host was attached?
 - v. Suppose a packet with an IP address 184.86.31.2 arrives at the router. Does it belong to the same subnet? Justify your answer.
- b) Suppose a 9000-byte IP packet is forwarded across a link with a 1500-byte MTU. How many fragments will be created? What are their lengths?



National University of Computer and Emerging Sciences, Lahore Campus

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Section:	All	Page(s):	2
Exam Type:	Sessional-2		

Student Name: Ayesha Javed

Roll No. L15-4522 Section: EE-2

- Instruction/Notes:
1. Write your Name and Roll Numbers in above space.
 2. Solve on Answer Sheets.

Question No. 1

CLO 03 Recognize the model of MAC layer and its performance parameters.

Marks: 5+5

- a) We have come across three generations of Carrier Sense Multiple Access protocols -- the original CSMA, CSMA/CD, and CSMA/CA. Based on your knowledge to these variants of CSMA MAC protocols, address the following questions:
- i. How does CSMA work in principle?
 - ii. What is the problem in CSMA that CSMA/CD is trying to resolve?
 - iii. What is the problem in CSMA/CD which CSMA/CA is trying to resolve?
- b) Suppose two nodes, A and B, are attached to opposite ends of an 800m cable, and that they each have one frame of 1,500 bits (including all headers and preambles) to send to each other. Both nodes attempt to transmit at time $t = 0$. Suppose there are four repeaters between A and B, each inserting a 20-bit delay. Assume the transmission rate is 100 Mbps, and CSMA/CD with backoff intervals of multiples of 512 bits is used. After the first collision, A draws $K = 0$ and B draws $K = 1$ in the exponential backoff protocol. Ignore the jam signal and the 96-bit time delay.
- i. What is the one-way propagation delay (including repeater delays) between A and B in seconds? Assume that the signal propagation speed is 2×10^8 m/s.
 - ii. At what time (in seconds) is A's packet completely delivered at B?

Question No. 2

CLO 04 Research the wireless networks and their applications

Marks: 4+3+3

Give short to-the-point answers to the following questions:

1. MACA solves both hidden terminal and exposed terminal problems while CSMA/CA solves only one of these problems. Which problem CSMA/CA does not solve and Why?
2. What is the purpose of varying Inter-frame Spacings in 802.11?
3. Why are there multiple Address fields in 802.11 frame instead of two (source MAC address and destination MAC address) fields?

Q.1 Suppose that host A is connected to a router R1, R1 is connected to another router, R2, and R2 is connected to host B. Suppose that a TCP message that contains 900 bytes of data and 20 bytes of TCP header is passed to the IP code at host A for delivery to host B. Show the **Total length**, **Identification**, **DF**, **MF**, and **Fragment offset** fields of the IP header in each packet transmitted over the three links. Assume that link A-R1 can support a maximum data link frame size of 1024 bytes including a 14-byte frame header, link R1-R2 can support a maximum data link frame size of 512 bytes, including a 8-byte frame header, and link R2-B can support a maximum data link frame size of 512 bytes including a 14-byte frame header. [20]

Q.2 (a) A large number of consecutive IP addresses are available starting at 202.67.0.0. Suppose that four organizations, A, B, C, and D, request 1500, 3000, 2000, and 16000 addresses respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation. [10]

(b) A router has the following (CIDR) entries in its routing table: [10]

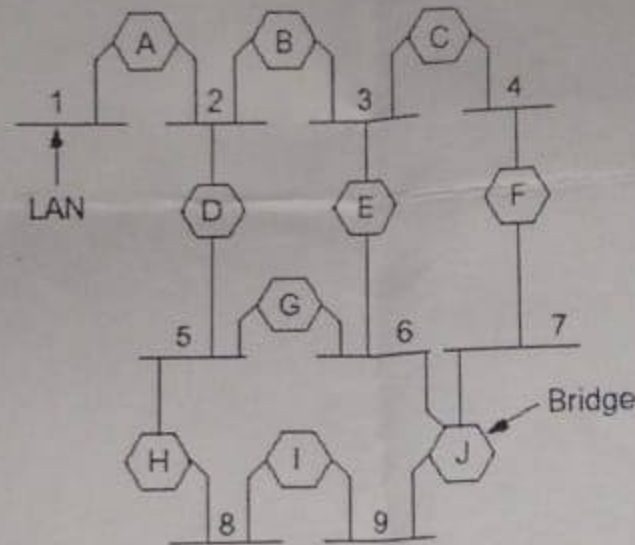
<u>Address/mask</u>	<u>Next Hop</u>
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Router 1
default	Router 2

For each of the following IP addresses, what does the router do if a packet with that destination address arrives? Show all the calculations if you want any credit for this question.

- (i) 135.46.63.10
- (ii) 135.46.57.14
- (iii) 135.46.52.2
- (iv) 192.53.40.7
- (v) 192.53.56.7

Q.3 A bit stream 10011101 is transmitted using the standard CRC method. The generator polynomial is $x^3 + 1$. Determine the actual bit stream transmitted. If the third bit from the left is inverted during transmission, show that this error is detected at the receiver. [20]

- Q.4** (a) One consequence of using a spanning tree to forward frames in an extended LAN is that some bridges may not participate at all in forwarding frames. **Redraw the following figure and identify such bridges and their links.** Is there any reason for keeping these bridges, even though they are not used for forwarding? [10]



- (b) Sketch the Manchester encoding on a classic Ethernet for the bit stream 0001110101. [10]

- Q.5** (a) Assume we need to design a Go-Back-N sliding window protocol for a network in which the bandwidth is 100 Mbps and the average distance between sender and receiver is 10,000 km. Assume the average packet size is 10^5 bits and the propagation speed in the media is 2×10^8 m/s. Find the size of the sender and receiver windows for **100% link utilization**, the number of bits in the sequence number, and an appropriate timeout value for the timer. [10]

- (b) A high quality speech signal has a bandwidth of 8 kHz. Suppose that the speech signal is to be quantized and then transmitted over a 28.8 kbps modem. What is the signal to noise ratio (in dB) of the received signal? What modem speed is needed if we require an SNR of 40 dB? [10]

National University of Computer and Emerging Sciences, Lahore Campus



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 Duration: 3 hrs
 Paper Date: 15-December-16
 Section: ALL
 Exam: **FINAL**

Course Code: EE317
 Semester: Fall 2016
 Total Marks: 100
 Weight: 50%
 Page(s): 2
 Roll No: 13L-4446

Instruction/Notes: Show all working/ steps required to get the final answer. Attempt all parts of a question together.

~~Q1~~ You are entrusted with the design of a network to interconnect a set of geographically distributed hosts within your corporation. You have decided to use a copper-based, switched network with an 8-point QAM encoding. The technology uses 5MHz signal bandwidth and 511 SNR at transmitter. What link bandwidth is possible according to Shannon's Law for this network? What modulation rate (baud) is necessary to obtain this bandwidth? (10)

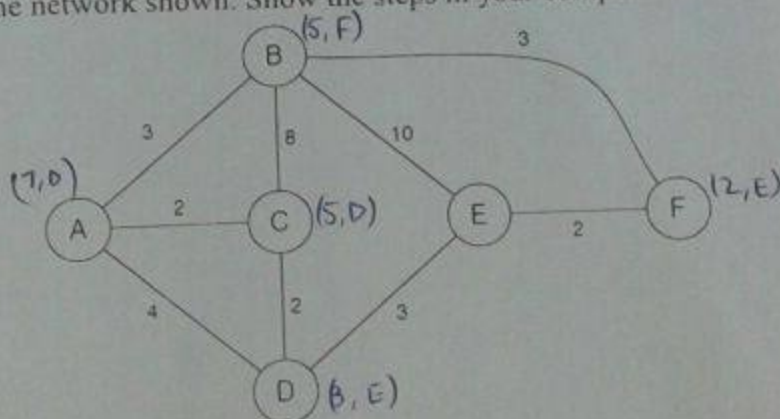
~~Q2~~ Frames of 1000 bits are sent over a 1-Mbps channel using a geostationary satellite whose propagation time from the earth is 270 msec. Acknowledgements are always piggybacked onto data frames. The headers are very short. Three-bit sequence numbers are used. What is the maximum achievable channel utilization for Stop-and-wait and Selective Repeat? (10)

The USB protocol uses a CRC with the polynomial $x^5 + x^2 + 1$. (20)

~~Q3~~ Calculate the CRC of the bit string 10011001. Assume bit string 1011101111000 is received (using the same generator G as in a), determine if error could be detected or not.

~~Q4~~ Consider a frame consisting of four bits. Assume that the probability of error is 10^{-3} , which is independent for each bit. What is the probability that the frame is received correctly?

~~Q5~~ Suppose that routing information is exchanged using link state routing protocol and the shortest paths are computed using Dijkstra's algorithm. Find the shortest path from E to all other nodes in the network shown. Show the steps in your computation. (10)



~~Q6~~ What are distance vectors in nodes E, D, and C? Note: you do not have to run the distance vector algorithm; compute distance vectors by inspection. (10)